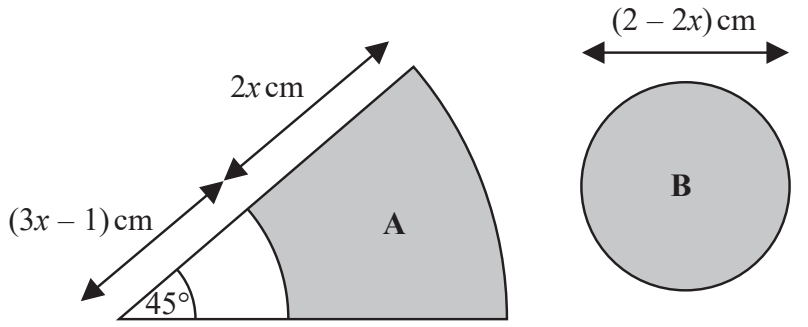
22 The diagram shows two shaded shapes, **A** and **B**.

Shape **A** is formed by removing a sector of a circle with radius $(3x - 1)$ cm from a sector of the circle with radius $(5x - 1)$ cm.
Shape **B** is a circle of diameter $(2 - 2x)$ cm.



The area of shape **A** is equal to the area of shape **B**.

Find the value of x.
You must show all your working.

Outer radius = (3x – 1) + 2x = 5x – 1.

Area A = π/8 [(5x – 1)² – (3x – 1)²]

= π/8(16x² – 4x); area B = π(1 – x)².

Equate, cancel π: 2x² – x/2 = 1 – 2x + x².

Multiply by 2: 2x² + 3x – 2 = 0.

(2x – 1)(x + 2) = 0; x = 1/2 or –2.

Only x = 1/2 gives valid positive lengths.

5 marks total
Sector annulus area
Expand both areas
Quadratic equation
Factorise
Valid x = 1/2

.....

(Total for Question 22 is 5 marks)

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